Faculty of Electronic Engineering, University of Niš



Seminar Paper

Topic: **Operation of a Raspberry Pi Microcomputer for Air Quality Measurement, Light Indication and Automatic Ventilation of a Space**

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Introduction

Motivation

Modern buildings often have poor air quality. Air quality is neither measured nor taken into account. Polluted air directly affects the health of the people (and other living beings) who stay there.

The Role of Embedded Systems (Raspberry Pi and IoT)

The development of microcontrollers and microcomputers (such as the Raspberry Pi) makes it possible to build inexpensive, fast and extremely reliable smart devices that can run 24/7 without a break.

Detection and Active Protection Concept (Brief Description of the Idea):

A system that connects a chemical sensor that reacts to changes in the air, converts that information into digital form, analyzes it using software on the Raspberry Pi, and then uses a feedback loop – if there is danger, the system simultaneously activates an alarm and a fan that physically clears the room.

Objective

The objective of this paper is the practical realization of a prototype of such a safety system, through the synergy of the Linux I2C subsystem, hardware components and the C programming language.

Assignment

The task is to build an air quality detection system based on a Raspberry Pi 4 B microcomputer using an MQ-2 sensor which, depending on the measured quality, will light the appropriate LED (green if the measured value is <= 45, yellow if it is 46 - 55 and red if it is >= 56) and, when needed, activate the buzzer (>= 56). The measurement results should also be visible on the LCD screen.

Required Components

- Raspberry Pi 4 B microcomputer + GPIO extension board + proto board (already connected)
- MQ-2 sensor
- PCF8591 AD/DA converter (8-bit)
- Active buzzer

- 16x2 LCD display
- 10kΩ potentiometer (for adjusting the screen brightness)
- Green, yellow and red LEDs
- 3 x 10kΩ resistors
- 3 x 1kΩ resistors
- 3 x BC547B transistors (BJT NPN)
- DFRobot 130 DC (comes with a fan and built-in drivers/transistors and allows direct connection)
- A smaller board (onto which the diodes, transistors and resistors will later be soldered)
- Soldering iron
- Solder wire
- An unknown number of wires/connectors (of various types; male-female and male-male)
- I/O devices for accessing the microcomputer (optional; for example, a keyboard alone may be enough, and possibly a screen so that you can see what is being typed; SSH access is possible if it is set up at the start during the configuration of the Raspberry Pi Imager):
    - Keyboard
    - Mouse
    - Screen

Theoretical Background

Raspberry Pi 4 B Microcomputer

An intuitive lightweight Linux distribution. Familiarity with the terminal (commands) is desirable for easier and faster use/navigation, although it is not crucial.

GPIO extension board

This is just a “dumb” component. It extends the Raspberry Pi’s ports.
It provides access to several GND and VCC/VOUT pins (5 and 3.3 V) and a handful of signal pins. There are 40 in total, each has its name/description next to it, and they are numbered 1-40, as follows:

- The top left one (as seen with the colored set of cables attached, taking that as the top position) is number 1 (labeled 3V3 on the board)
- Then the one across from it (labeled 5V0) is number 2
- And so on downwards (1, 3, 5, …, 39 on the left side and 2, 4, 6, …, 40 on the right)

Proto Board

It is necessary to know how a proto board works.

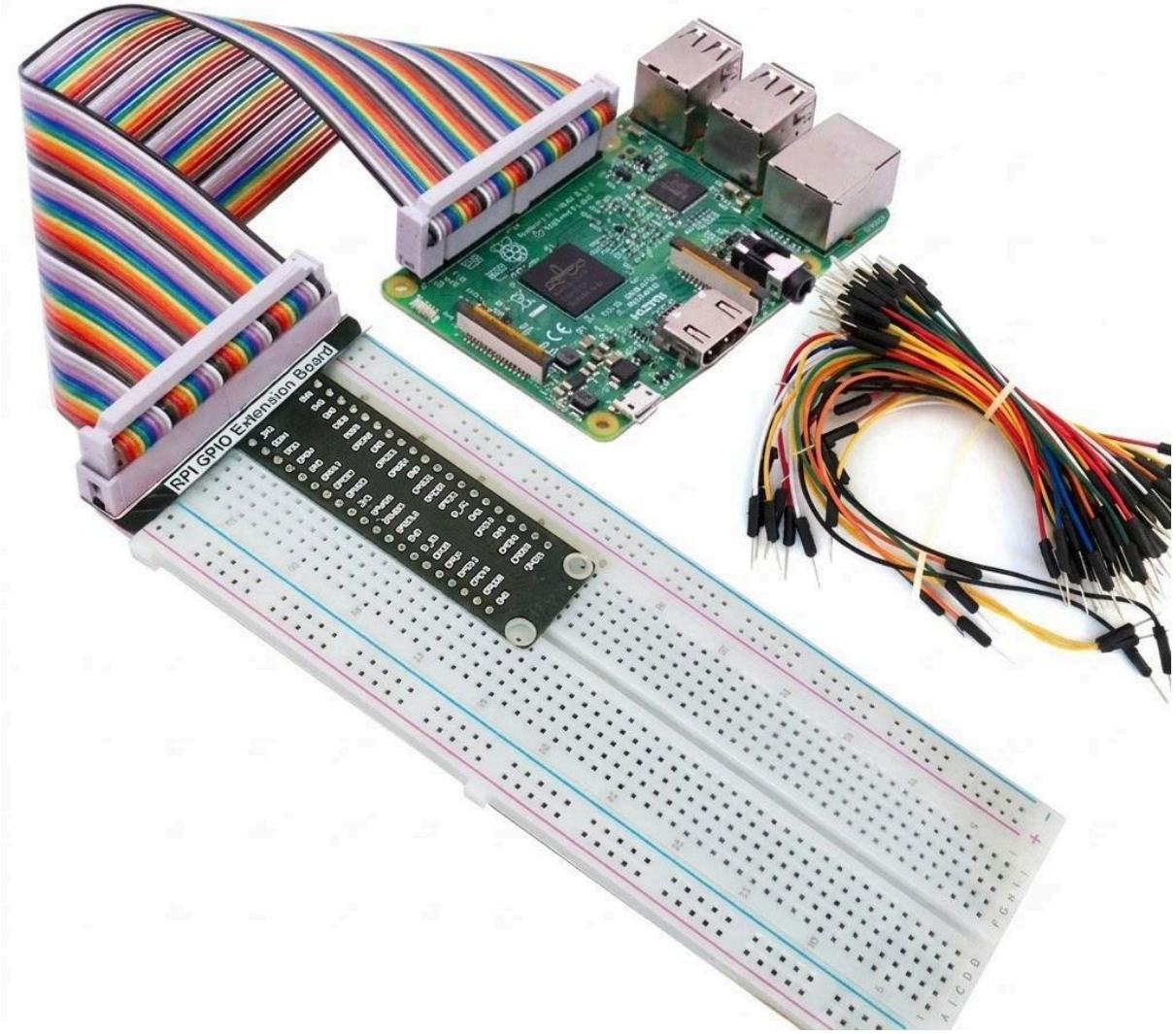


Figure 1. a) Raspberry Pi 4 B + GPIO extension board + cables + proto board **[1]**

MQ-2 Sensor

The MQ2 is one of the most popular components of the MQ series for detecting smoke and flammable gases, which is why it is widely used in home, office and industrial safety systems. This semiconductor metal-oxide sensor (MOS), also known as a chemiresistor, works on the principle of a change in the electrical resistance of the sensing material. In contact with gas, its resistance decreases in proportion to the growth in particle concentration. It is intended for the detection of liquefied petroleum gas (LPG), smoke, alcohol, propane, hydrogen, methane and carbon monoxide, with a measuring range from 200 to 10000 ppm **[2]**.

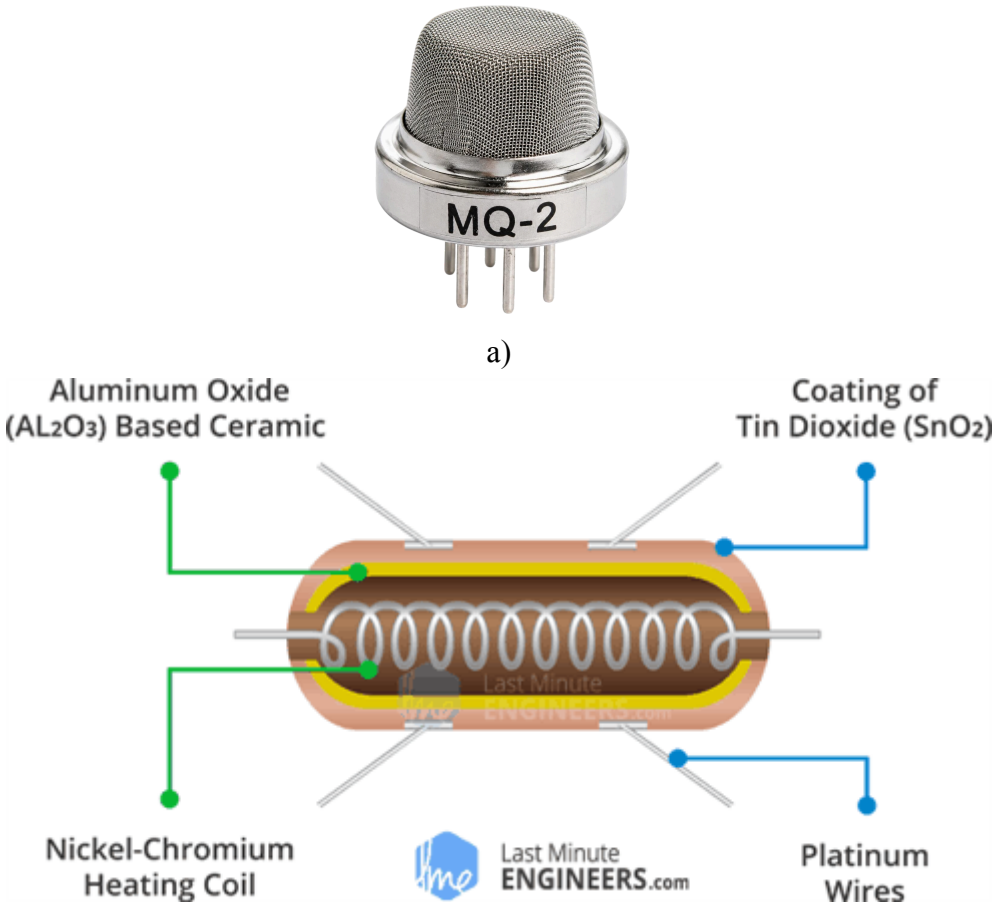


Figure 2. a) Appearance of the MQ2 sensor **[11]**, b) Internal structure of the sensor **[12]**

Internal Structure of the MQ2 Sensor

A protective double-layer stainless steel mesh encloses the sensing element (Figure 2 a)) and is fixed to the body of the component with a clamping ring. Its primary role is safety – it prevents an explosion if the internal heater comes into contact with flammable mixtures, while at the same time acting as a filter that lets only gases through, protecting the interior from solid particles.

In the central part, protected by the mesh, are the heating and sensing elements:

- **Heating element** is made in the form of a nichrome (NiCr) coil on an aluminum-oxide ceramic substrate (Al_2O_3).
- **Sensing element** consists of platinum (Pt) conductive wires and an active coating of tin dioxide (SnO_2).

It is precisely tin dioxide that is the key component of this assembly, because its chemical properties enable it to react to and detect the gases present in the environment **[2]**.

Operating Principle of the MQ2 Sensor

The detection process begins with heating the tin dioxide using the internal heater, which encourages the binding of oxygen from the environment. In clean air, donor electrons from the tin dioxide react with oxygen atoms absorbed on the surface of the material. This binding of electrons blocks the passage of electric current, which manifests as a high resistance of the sensing element.

Conversely, when flammable or toxic gases appear in the air, their molecules interact with the absorbed oxygen, which reduces its surface density. As a result of this process, the electrons inside the tin dioxide remain free, the conductivity of the material increases, and the overall resistance of the sensor decreases **[2]**.

MQ2 Sensor Module

In practice, the MQ2 sensor is integrated onto a dedicated printed module that considerably simplifies its use and provides two types of output signals (Figure 3 a)):

- **Digital output** – generates a binary signal that gives the user quick information on whether flammable gases are present.
- **Analog output** – delivers a voltage signal whose value is directly proportional to the amount of smoke or gas in the space (a rise in concentration causes a proportional rise in output voltage). This output is used for analysis and calculation of the current state of the air.

A high-precision comparator integrated circuit LM393, integrated on the module itself, is used to digitize the analog signal. The detection threshold, i.e. the minimum gas concentration to which the system will react, is configured manually using a 10kΩ trimmer potentiometer (Figure 3 b)). In addition to the components mentioned, the board also carries two decoupling capacitors intended to suppress electrical noise, as well as a pair of signal LEDs that display the power status and indicate operation.

It is important to stress that most sensors of this group are of the multi-gas type. Although they detect changes in the total gas concentration very effectively through variations in output voltage, they have no ability for selective segregation. This means that the MQ2 sensor cannot identify the exact type of gas present or its absolute individual concentration, but is primarily intended for detecting general changes in air quality **[2]**.



a) b)

Figure 3. MQ2 gas sensor module. a) model with pin layout **[13]**, b) appearance of the module in reality **[14]**

Pin Layout of the MQ2 Sensor Module

The labels and layout of the connectors on the module are shown in Figure P6-2. a). The module has a total of four functional pins:

- **VCC** – Power supply connection of the module, connected to the 5 V source from the microcomputer.
- **GND** – Ground of the system, connected to the corresponding GND potential of the microcomputer.
- **D0** – Digital output that delivers a binary signal (information) about the detection of flammable gases in the environment.
- **A0** A0 – Analog output through which a voltage signal proportional to the current concentration of smoke or gas is sent [2].

Calibration of the MQ2 Gas Sensor

Calibration of the MQ2 sensor is carried out by placing the component in the immediate vicinity of the target gas, after which the trimmer potentiometer is adjusted until the red signal LED lights up. The direction of rotation of the potentiometer directly affects the detection parameters: turning it clockwise raises the sensitivity level, while turning it counterclockwise reduces the sensitivity to a given gas.

The LM393 integrated circuit is responsible for monitoring changes. This comparator continuously compares the voltage from the analog output (A0) with a predefined reference threshold set using the potentiometer. At the moment the analog signal value exceeds that threshold, the digital output (D0) switches to a high logic state (high logic level), which simultaneously causes the red LED to light up **[2]**.

Preheating Time of the MQ2 Gas Sensor

For correct operation and high measurement accuracy, a preheating period, i.e. a stabilization time, is defined for this group of components. For the MQ2 sensor this process lasts 24 hours, which implies that when first put into operation the device must run continuously for that entire period. Meeting this condition is necessary so that the characteristics of the sensor correspond to the specifications given in its technical documentation (datasheet). In addition, besides the initial stabilization, regular calibration is mandatory to achieve maximum precision, as well as software or hardware compensation for external environmental factors, above all temperature and air humidity **[2]**.

PCF8591 8-bit A/D and D/A Converter

General Description

The PCF8591 is a single-chip, single-supply, low-power CMOS data acquisition device. It contains four analog inputs, one analog output and a serial I^2C interface. Three address pins – A0, A1 and A2 – are used for programming the hardware address, which allows up to eight devices to be connected to the same I^2C bus without additional hardware. The address, control and data are transferred serially between the devices over the two-line bidirectional I^2C bus **[3]**.

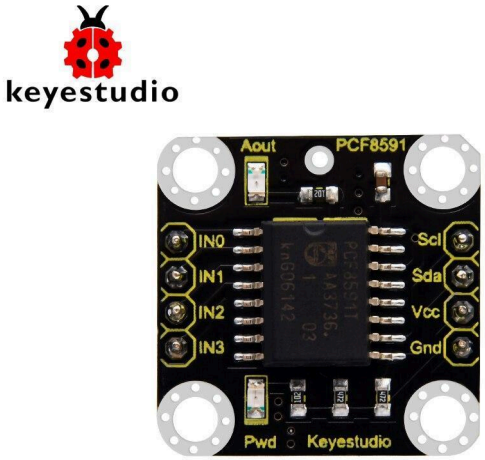


Figure 4. Appearance of the PCF8591 converter

The functions of this device include:

- Analog input multiplexing.
- An integrated on-chip “track and hold” function.
- 8-bit analog-to-digital (A/D) conversion.

- 8-bit digital-to-analog (D/A) conversion.
- The maximum conversion rate is determined by the maximum speed of the I^2C bus **[3]**.

Features and Advantages

- Single power supply: operating voltage from 2.5 V to 6.0 V.
- Low standby current.
- Serial input and output via the I^2C bus.
- Selection of the I^2C address using 3 hardware pins.
- 4 analog inputs that can be configured as single-ended or differential inputs.
- Automatic incrementing when selecting a channel (auto-increment).
- Analog voltage range from V_{SS} to V_{DD}.
- A/D conversion using the successive approximation method **[3]**.

Pin Description

The device comes in DIP16 (PCF8591P) or SO16 (PCF8591T) packages with the following pin layout:

- **Pins 1, 2, 3, 4 (AIN0, AIN1, AIN2, AIN3):** Analog inputs for the A/D converter.
- **Pins 5, 6, 7 (A0, A1, A2):** Hardware slave addresses.
- **Pin 8 (V_{ss}):** Power ground.
- **Pin 9 (SDA):** Serial data transfer for the I^2C bus.
- **Pin 10 (SCL):** Serial clock for the I^2C bus.
- **Pin 11 (OSC):** Oscillator input/output.
- **Pin 12 (EXT):** External/internal switch for the oscillator input (connected to V_{ss} when the internal oscillator is used).
- **Pin 13 (AGND):** Analog ground.
- **Pin 14 (V_{REF}):** Reference voltage input for the conversion.
- **Pin 15 (AOUT):** Analog output of the D/A converter.
- **Pin 16 (V_{DD}):** Supply voltage **[3]**.

Functional Description

- **Addressing:** The first byte sent after the START condition is the I^2C address. It consists of a fixed part (1001) and a programmable part defined by the states on pins A2, A1 and A0. The last bit (bit 0) determines the read or write operation ($R/\overline{W}$) **[3]**.
- **Control byte:** The second byte is stored in the control register and governs the functions of the device. The upper nibble is used to enable the analog output and configure the analog inputs, while the lower nibble selects the active analog channel. If the “auto-increment” flag is set, the channel number is automatically incremented after each A/D conversion **[3]**.
- **D/A conversion:** The third byte sent to the device is stored in the DAC data register and converted into an analog voltage by an internal 256-step resistor ladder. The output voltage is buffered by a unity-gain amplifier **[3]**.

- **A/D conversion:** Uses the successive approximation technique. The conversion cycle is triggered on the falling edge of the acknowledge clock pulse after a valid read address is sent, and is carried out during the transfer of the result of the previous conversion. The result of conversion of differential inputs is encoded as an 8-bit two's complement **[3]**.

Buzzer. Active Buzzer

Buzzer

Buzzers (sounders or acoustic signaling devices) are miniature electroacoustic transducers which, based on their physical principle of operation, are classified into two main groups:

1. **Piezoelectric buzzers** – generate an acoustic signal under the influence of an applied voltage. Electrically, this type of buzzer is modeled as a capacitor **[2]**.
2. **Electromagnetic buzzers** – produce sound by current flowing through windings, and in electrical circuits are represented as a series connection of an inductance (coil) and a resistor **[2]**.

Construction and operating principle of a piezoelectric buzzer This component consists of a thin disc made of piezoelectric material, whose side surfaces are coated with electrodes for applying the excitation signal. The edges of the disc itself are fixed to the housing of the device. When a voltage is established between the electrodes, mechanical deformation and micro-displacement of the piezoelectric material occurs. These rapid oscillations of the disc within the acoustic chamber of the housing create sound waves.

According to the way the excitation is generated, piezoelectric buzzers are divided into active and passive:

- **Active buzzers** – have an integrated internal oscillator (driver) circuit. To activate them they require only a direct voltage (DC), which they convert internally into a continuous or pulsed sound signal of fixed frequency. They behave like classic sound indicators and start emitting sound immediately once power is applied. Because of their simplicity of use, they find wide implementation in computer peripherals, printers, alarm systems, automotive electronics, timers and similar devices **[2]**.
- **Passive buzzers** – do not contain their own oscillator, so they require an external alternating signal to operate. They are most often driven by a train of square pulses (PWM signal) in the frequency range from 2 kHz to 5 kHz. They work on the principle of a miniature loudspeaker, where the frequency and shape of the input alternating signal directly dictate the characteristics of the reproduced sound **[2]**.

A characteristic of piezoelectric buzzers is their simple and robust construction. However, they reproduce a relatively monotonous and piercing tone of fixed timbre that cannot be modulated, which is why they are primarily used to implement alarm functions. When

connected to development systems such as Arduino boards, these elements serve exclusively for audible signaling of events – for example, when a motion sensor registers presence or when some measured physical quantity exceeds a critical current threshold **[2]**.

While the distinction between piezoelectric and electromagnetic buzzers is obvious from their physical and electrical properties, visually telling active from passive piezoelectric variants is often a problem in practice, which creates the need to define clear methods for identifying them **[2]**.

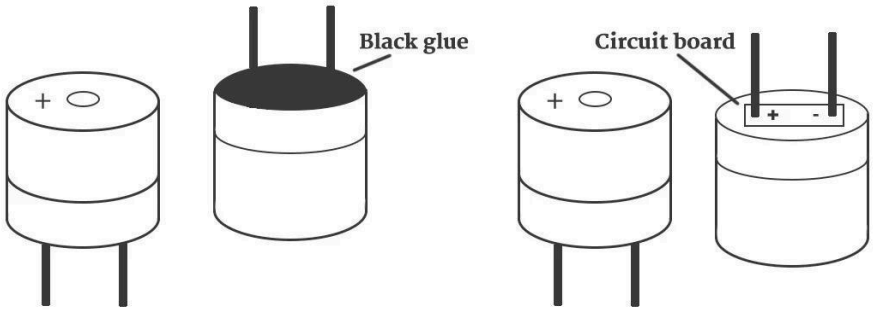
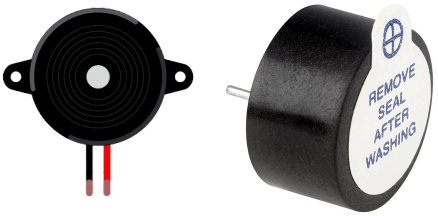


Figure 5. Difference between an active (left) and a passive buzzer (right) **[15]**



a) b)

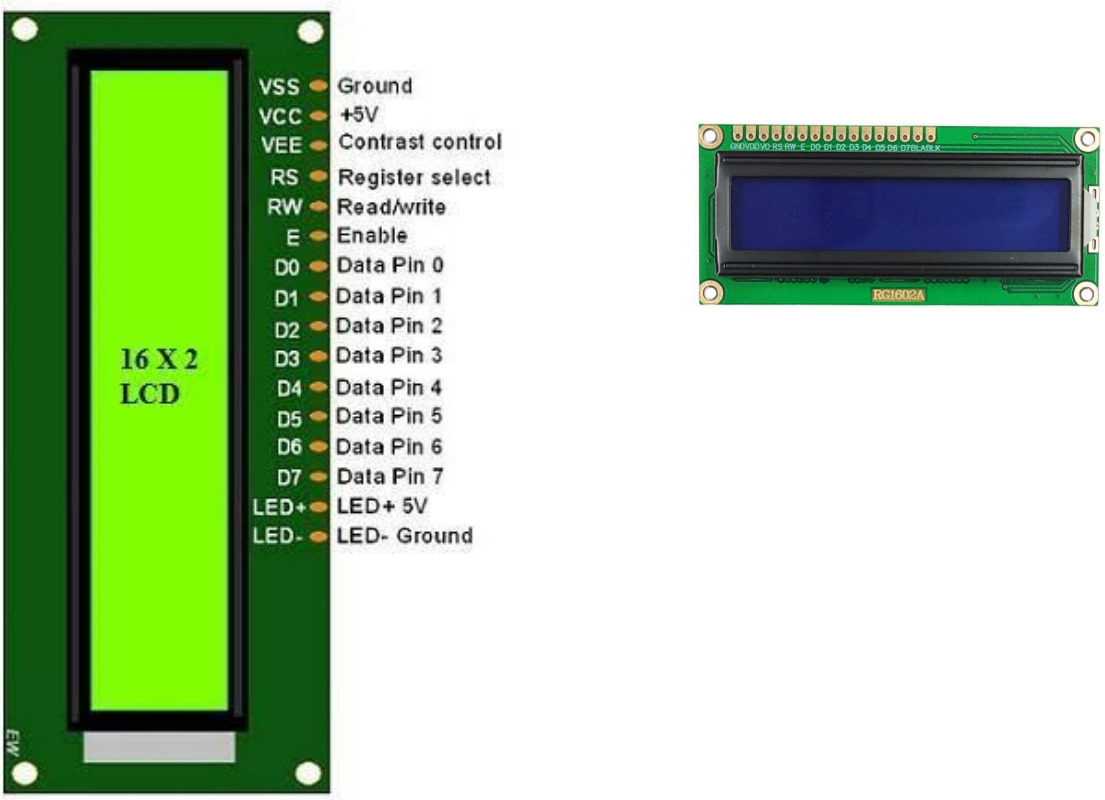
Figure 6. Active buzzer: a) model with pin layout **[15]**, b) appearance of the buzzer in reality **[16]**

To precisely determine the type of a piezoelectric buzzer in practice, two basic methods of visual identification are applied:

- **Geometric dimensions (housing height)** – Active and passive variants differ slightly in their dimensions; the housing of an active buzzer is 9 mm high, while the body of a passive buzzer is slightly lower, at 8 mm.
- **Material and structure of the underside** – When the components are positioned with their leads (pins) pointing upward, a black surface sealed with vinyl resin can be seen on the underside of the active buzzer. In contrast, with the passive version the green printed circuit board with exposed contours of the internal connections remains visible (Figure 5) **[2]**.

16×2 LCD Display

The measurement results are visualized using a 16×2 LCD display, which shows a value in the range from 0 to 255 representing the digital equivalent of the analog output voltage from the sensor **[2]**. This data is generated at the output of the integrated A/D converter within the PCF8591 module, and is then forwarded via the I2C bus to the microcomputer for further processing **[2]**. Cause and effect: every increase in the concentration of harmful gases in the environment causes a proportional rise in voltage at the sensor output itself, which directly results in a higher numerical value at the A/D converter and a corresponding display on the LCD screen **[2]**.



a) b)

Figure 7. 16x2 LCD display: a) model with pin layout **[18]**, b) appearance of the display in reality **[19].**

When configuring the 16×2 LCD display, the 4-bit mode of operation is used, as a result of which the data pins D0–D3 remain inactive (unconnected). A graphic representation of the model with the pin layout, as well as the physical appearance of the component itself, are given in the corresponding diagrams of the project.

A trimmer potentiometer with a resistance of 10kΩ is used to adjust the display contrast intensity; its schematic symbol and real appearance are illustrated in the accompanying figures. This potentiometer has three functional leads:

- **VCC** – Connection for the 5 V power supply.
- **GND** – Connection for the reference potential (ground).
- **VOUT** – Output pin for LCD contrast control, connected directly to the V0 pin of the display.

On the software side, to abstract complex low-level commands and speed up data output, most programming languages offer dedicated implemented libraries. The specific software implementation for this project relies on an appropriate library for managing LCD modules, which enables simple control of the display **[2]**.



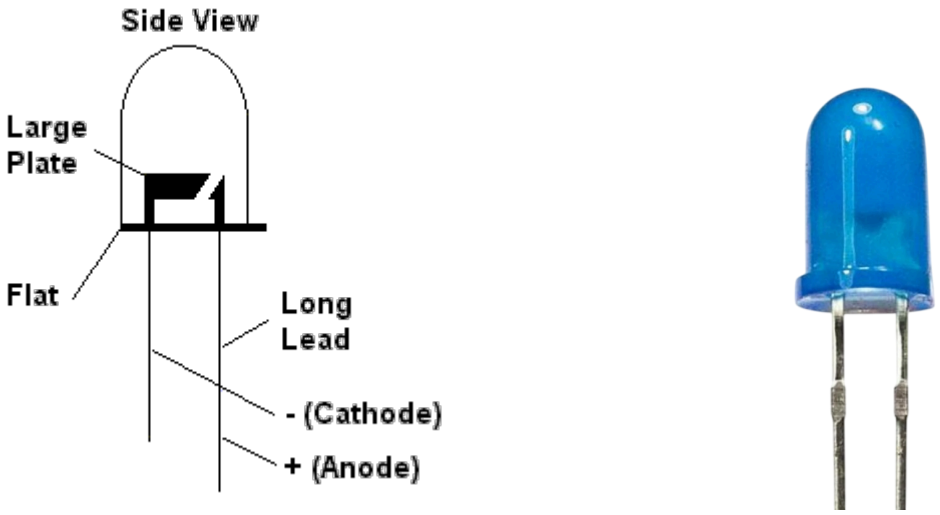
a) b)

Figure 8. Potentiometer for display contrast control: a) model with pin layout **[20]**, b) appearance of the potentiometer in reality **[21]**.

LED Diode

A light-emitting diode, or LED (Figure 9), is a semiconductor component that emits light when current passes through it in the forward-bias direction – from the anode (positive lead, recognizable as the longer pin) to the cathode (negative lead, i.e. the shorter pin) **[2]**. In terms of internal structure, this component is a PN junction formed by joining a P-type semiconductor (where holes dominate as the majority charge carriers) and an N-type semiconductor (with an increased concentration of free electrons).

When an appropriate excitation voltage is applied to the leads (so that the potential of the anode is higher than that of the cathode), the diode becomes conductive. This enables the process of recombination of electrons and holes in the zone of the PN junction itself. During this process, electrons drop to a lower energy level, and the excess energy is released in the form of light quanta – photons. The basic functional property of a diode is unidirectional conductivity, meaning it allows current to flow only in the forward-bias direction, while in the reverse direction it acts as a blockade **[2]**.



a) b)

Figure 9. LED: a) model with pin layout **[22]**, b) appearance in reality **[23]**.

To prevent damage due to excessive current flow, a current-limiting resistor must always be connected in series with an LED in electrical circuits, which means this component is never connected directly to power sources **[2]**. Although a 1 kΩ resistor is standardly used with a 12 V supply, the appropriate resistance value can be calculated analytically for any voltage value **[2]**. In this particular laboratory exercise, a series resistor of 220 Ω is used to protect the diode. Regardless of the components chosen, it is crucial to ensure correct polarity during mounting – the anode must be placed at a higher potential than the cathode in order to establish the conductive state **[2]**.

A characteristic of light-emitting diodes is a high degree of efficiency in converting electrical energy into light, with minimal losses in the form of unwanted heat dissipation. The spectrum of the emitted light (from the visible or invisible range, i.e. the color) depends directly on the chemical composition and type of the semiconductor material used in making the PN junction:

- **Red color** – covers wavelengths from 610 nm to 760 nm, and is achieved using semiconductors such as aluminum gallium arsenide (AlGaAs) or gallium arsenide phosphide (GaAsP) **[2]**.
- **Yellow color** – covers the range from 580 nm to 595 nm, where aluminum gallium indium phosphide (AlGaInP) or gallium phosphide (GaP) are used as active materials **[2]**.
- **Green color** – corresponds to wavelengths from 500 nm to 560 nm, which is achieved by structuring junctions of indium gallium nitride (InGaN) or gallium nitride (GaN) **[2]**.

Resistor

A resistor (Figure 10) is a component that limits (reduces) the current. It is a non-polarized component (the orientation of connection does not matter).



Figure 10. A typical resistor **[4]**.

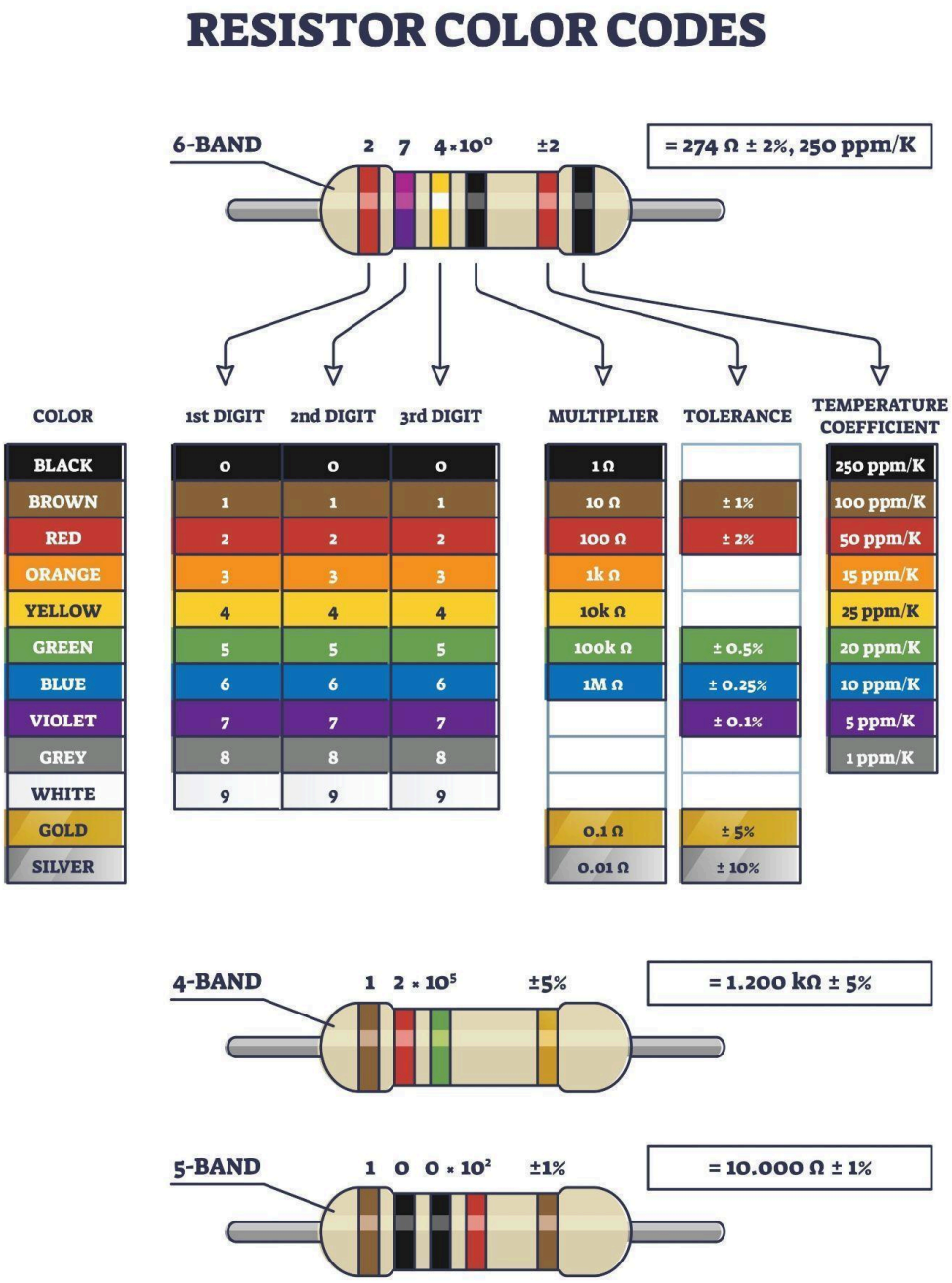


Figure 11. The so-called color code **[5]**.

BC547B Transistor

The BC547B (Figure 12) is a general-purpose bipolar NPN transistor, and its operation is tied to the ability to switch between the reverse-biased and forward-biased states, which allows it to fulfill roles in various electronic applications. When the base pin is grounded, the transistor

remains in the reverse-biased state, which results in an open circuit between the collector and the emitter **[6]**.



Figure 12. Appearance of the BC547B transistor **[6]**.

What Is the BC547B?

The **BC547B** is a commonly used NPN transistor, notable for its gain, which ranges from 110 to 800. This gain highlights its strong amplification capability, which is an important detail for many electronic circuits. The collector pin of this transistor supports a maximum current of 100 mA, which indicates that it is not suitable for high-load applications. For proper biasing, the current at the base pin (I_B) must not exceed 5 mA **[6]**.

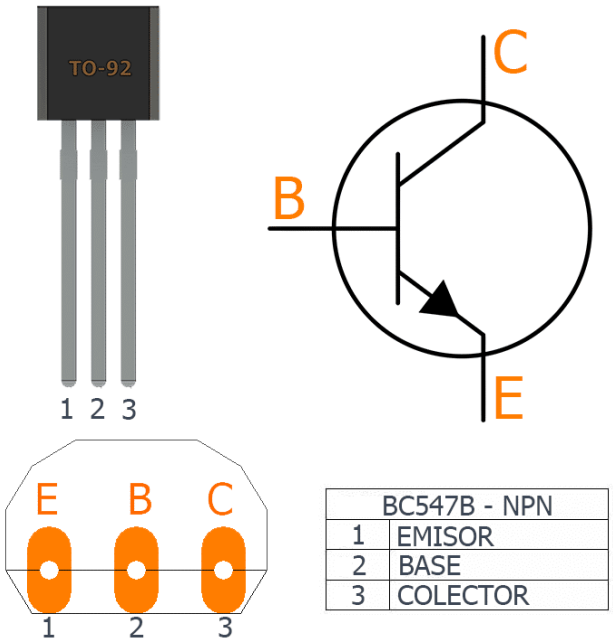
Under full bias, this device can handle a current of up to 100 mA between the collector and the emitter. In the saturation region, the voltage between the collector and emitter (V_{CE}) or between the base and emitter (V_{BE}) varies from 200 to 900 mV. When the base current is switched off, the transistor enters the cut-off region, with a base-emitter voltage of approximately 660 mV, which effectively stops the flow of current. Understanding these operating parameters is necessary for practical application. For example, when designing a low-power amplifier, the selection of components such as the BC547B is needed in order to ensure optimal performance and device longevity by operating within safe limits.

Alternatives to the BC547B [6]

- BC547BBU
- BC547BTF
- BC547BTA
- BC549
- BC636
- BC639

- 2N2222 (TO-92, TO-18)
- 2N2369
- 2N3055
- 2N3904
- 2N3906
- 2SC5200

Configuration



Pin number	Pin name	Description
1	**Collector**	**Current enters through the collector**
2	**Base**	**Controls the biasing of the transistor**
3	**Emitter**	**Current leaves through the emitter**

Table 1. Pin configuration of the BC547B **[6]**.

Characteristics

BC547B is a bipolar junction transistor (**BJT**) of the NPN type, highly valued for its ability to amplify current. Its versatile nature makes it an indispensable component in amplification and switching scenarios. The NPN structure effectively allows electrons to flow from the emitter to the collector, which is achieved with a small input current at the base. This behavior is used in numerous electronic circuits **[6]**.

The BC547B is characterized by a high DC current gain (h_{FE} i.e. DC current gain) of up to 800. This allows the transistor to multiply the input current, which is a great advantage when amplifying low-power signals. Small variations in the input signal can thus be converted into considerably larger output signals. With the ability to handle a continuous collector current of up to 100 mA, the BC547B can drive moderate power loads. This capacity proves useful in various small-scale devices, such as driving LEDs or acting as a switch in various circuit modules. Historical data from practical applications suggest that keeping the current below this threshold increases the stability and longevity of the component **[6]**.

The rated emitter-base voltage of 6 V represents the maximum permitted voltage across the emitter-base junction. Observing this limit protects against potential damage, ensuring reliable operation in circuits where precise voltage control is required. The maximum base current of 5 mA is an integral factor for safe operation of the transistor. This parameter is crucial for designing biasing circuits, which set the operating point of the transistor. Exceeding this current could overload the junction, reduce performance and potentially cause irreversible damage **[6]**.

The BC547B is enclosed in a **TO-92** package, known for its compactness and wide use. This design makes handling, soldering and mounting on a printed circuit board (PCB) simple. The robustness of the TO-92 package and efficient heat dissipation further contribute to the reliability of the BC547B in various applications. Finally, the BC547B impresses with its high gain, moderate current-handling capabilities and solid physical package. These characteristics show not only its versatility, but also its reliable performance in multiple applications **[6]**.

Applications

Driver Modules and Control Circuits

The **BC547B** transistor often finds its place in driver modules, such as relay drivers and LED drivers. In a relay driver circuit, the transistor acts as a switch that handles low-current signals in order to control larger currents, creating a protective barrier between the controlling device (controller) and the relay. For example, home automation systems benefit greatly from this application, as it ensures smooth operation of relays that control various electrical appliances. In LED drivers, BC547B transistors regulate the current flowing through the LEDs, providing uniform illumination and extending the lifespan of the LEDs themselves **[6]**.

Amplifier Modules

In amplifier modules, the BC547B transistor plays a role both in audio amplification and in signal amplification. Audio amplifiers use these transistors to amplify low-power audio signals, which proves to be an excellent solution in devices ranging from hearing aids to high-fidelity (Hi-Fi) audio systems. By increasing signal strength without introducing distortion, BC547B transistors guarantee the clarity and fidelity of the output sound. Signal amplifiers boost weak signals to a functional level. This becomes particularly important in radio transmission, where signal integrity must be preserved over great distances **[6]**.

Darlington Pair Configurations

In Darlington pair configurations, two BC547B transistors are paired to provide extremely high current gain. This arrangement is especially useful in scenarios that require high current amplification with minimal input current. For example, in industrial automation, Darlington pairs enable precise control over actuators and sensors, thereby improving operational efficiency and the response speed of the system **[6]**.

Use as a Switch

When used as a switch, the **BC547B** transistor switches between the saturation region (acting as a closed switch) and the cut-off region (acting as an open switch). Understanding these operating modes is the key to achieving smooth functionality. In the saturation region, both junctions — base-emitter and base-collector — are forward biased, which allows current to flow from the collector to the emitter. Conversely, in the cut-off region, both junctions are reverse biased, and no current flows through the collector-emitter path **[6]**.

Proper biasing of the base current ensures that the transistor transitions efficiently between these two states. The base current (I_B) should be limited and must not exceed 5 mA in order to prevent damage to the transistor. A series resistor (R_B) is used to maintain this limit, which protects the transistor and stabilizes its switching operation. The value of this resistor R_B can be calculated using the formula **[6]**:

$$R_B = V_{BE} / I_B$$

For the BC547B, the voltage V_{BE} is about 0.7 V. Therefore, for a desired base current of 1 mA, the series resistor should have a value of approximately 0.7 kΩ **[6]**.

Proper selection of the resistor and accurate calculations of the base current are important in practical applications. Addressing challenges such as voltage drops and maintaining temperature stability plays a large role in the reliability of performance. Implementing a heat sink or adequate ventilation can improve the longevity of the device and the stability of operation. BC547B transistors are often used in simple control circuits, relays and low-power switching applications. For example, in a relay driver circuit, the BC547B can switch the current going to the relay coil, enabling the control of higher-power devices using relatively

low control voltages. This approach emphasizes the importance of proper component selection and careful circuit design, which ensures reliable operation in different use cases **[6]**.

Use as an Amplifier

In the active region, the **BC547B** transistor effortlessly increases power, voltage and current. This dynamic operation comes to life through various configurations that are widely accepted in electronic circuits. The primary setups for the BC547B transistor include: the common-emitter amplifier, the common-collector amplifier and the common-base amplifier. Among them, the common-emitter amplifier enjoys great popularity in numerous applications due to its outstanding voltage amplification capabilities **[6]**.

The **common-emitter** setup is used extensively because of its superior voltage amplification qualities. It is known for inverting the input signal, which makes it a fundamental component in amplifier circuits. A clear example can be found in audio amplification, where this configuration significantly increases signal strength, effectively driving loudspeakers. The **common-collector**amplifier, also known as the emitter follower, is predominantly used for impedance matching. Although its voltage gain is around unity, it stands out in providing current gain. This setup is often used in buffer (isolation) stages for connecting high impedance to low-impedance loads, thereby preserving signal integrity **[6]**.

The **common-base**configuration, although less common, stands out in high-frequency applications due to its low input impedance. It is used in RF (radio-frequency) amplification where such characteristics are invaluable. In the operation of these setups, understanding the DC current gain (FE) is very important. This parameter, crucial for the amplification potential of the transistor, is calculated by the formula **[6]**:

$$h_{FE} = I_C / I_B$$

Calculations and adjustments based on h_{FE} ensure that the transistor operates within its optimal range, enabling reliable amplifier performance. Biasing networks are often stabilized to mitigate variations in the h_{FE} value between different transistors, thereby guaranteeing uniform performance. Applications also require thermal management to prevent changes in h_{FE} due to temperature fluctuations, which shows the importance of thermal considerations in effective circuit design **[6]**.

A deep understanding of these configurations and their applications brings significant advantages in the design of electronic circuits. The adaptability of the BC547B transistor as an amplifier, visible through applications ranging from audio systems to RF circuits, highlights its broad usability. Integrating these configurations into practical solutions can significantly improve the functionality and efficiency of electronic systems. The different amplification configurations of the BC547B transistor offer key building blocks in electronics, with each fulfilling specific roles in order to optimize circuit performance **[6]**.

Specifications

Type	Parameter
Lifecycle status	ACTIVE (Updated 6 days ago)
Contact plating/coating	Copper, silver, tin
Mounting type	Through Hole
Number of pins	3
Weight	200.005886 mg
Collector-emitter breakdown voltage	45 V
Number of elements	1
Operating temperature	150°C TJ
Published	2005
Moisture sensitivity level (MSL)	1 (Unlimited)
Minimum operating temperature	-65°C

Maximum power dissipation	500 mV
Maximum operating temperature	150°C
Rated DC voltage	45 V
Rated current	100 mA
Frequency	300 MHz
Polarity	NPN
Element configuration	Single
Power dissipation	500 mV
Gain bandwidth product	100 MHz
Collector-emitter voltage (VCEO)	45 V
Collector cut-off current (Max.)	15 nA ICBO
Collector-emitter breakdown voltage (Max.)	45 V

Maximum breakdown voltage	45 V
Collector-base voltage (VCBO)	50 V
Height	4.58 mm
Width	3.86 mm
Radiation hardening	No
Lead free	Lead free

Table 2. Specifications of the BC547B **[6]**.

DFRobot 130 DC Motor

Introduction

Gravity: 130 DC Motor Module (Figure 15) combines a classic size-130 brushed DC motor with an integrated transistor control circuit, creating a compact actuator suitable for rapid prototyping and embedded-systems projects involving motion. The integrated switching stage enables direct digital control from popular development platforms such as Arduino, without the need for a separate motor driver board. The standardized Gravity 3-pin connector simplifies wiring and improves reliability during experimentation. This small DC motor unit supports PWM-based (pulse-width modulation) speed control, which makes the module suitable for robotic mechanisms, airflow experiments and interactive hardware assemblies where quick installation and stable power switching are important **[8]**.



Figure 13. Appearance of the DFRobot 130 DC motor **[7]**.

Integrated Transistor Driver for Direct Control

The integrated transistor driver circuit allows direct switching from a digital pin of the microcomputer, which eliminates the need for an external motor driver module in many low-power applications. The built-in control stage simplifies circuit design and reduces the component count during rapid prototyping. PWM signals effectively regulate the rotation speed, enabling precise control for small mechanical systems such as mini fans, robotic actuators or experimental moving assemblies **[8]**.

Gravity 3-Pin Interface for Rapid Prototyping

The standardized Gravity 3-pin digital interface integrates the power, ground and signal connections into a single plug-and-play cable. This connector design minimizes wiring complexity and reduces the chance of connection errors during development. The compact actuator board integrates easily with Gravity-compatible controllers and expansion boards (shields), supporting a clean layout in educational electronics kits, embedded-system development environments and rapid hardware demonstrations **[8]**.

PWM Speed Control for Flexible Motion Output

PWM control makes it possible to adjust the motor speed over a wide operating range using simple commands in the firmware (code). This capability enables adjusting airflow in cooling experiments, rotary motion in light robotic mechanisms, as well as variable-speed

demonstrations in school electronics projects. Stable switching through the built-in transistor stage ensures reliable performance during continuous operation **[8]**.

Common application scenarios include small robotic mechanisms, “do-it-yourself” (DIY) fan control systems, educational electronics demonstrations, airflow testing setups, as well as interactive installations based on motion. The compact form factor, integrated switching circuit and Gravity interface design make this module exceptionally suitable for STEM learning kits, rapid prototyping environments and beginner-friendly embedded hardware development **[8]**.

Features

- **Integrated transistor driver** eliminates the need for an external motor driver.
- **Supports PWM speed control** for adjusting the rotation speed.
- **Standard Gravity 3-pin interface** for quick plug-and-play wiring.
- **Compact module** suitable for robotics and “do-it-yourself” (DIY) projects.
- **Compatible with common development boards** with microcomputers **[8]**.

Applications

- Small robotic motion systems
- “Do-it-yourself” (DIY) experiments with fans and airflow
- Interactive hardware installations
- STEM education electronics demonstrations
- Rapid prototyping of motor-driven mechanisms **[8]**

Specification

- **Operating voltage:** 5 V
- **Interface type:** Gravity 3-pin digital sensor interface
- **Dimensions:** 52 × 27 mm (2 × 1 in)
- **Weight:** 40 g **[8]**

Package Contents:

- Gravity: 130 DC Motor ×1
- Fan module ×1
- Gravity digital cable ×1

Soldering Iron and Solder Wire

Soldering iron (Figure 14) is an electric or gas tool used to melt solder material (most often solder wire made of tin and lead, or lead-free; Figure 15) in order to create a firm electrical and mechanical connection between two components or conductors. It is an indispensable tool in electronics, electrical engineering, and in various “do-it-yourself” (DIY) projects.



Figure 14. Soldering iron with a stand and a dedicated sponge **[9]** (the image is for demonstration purposes).

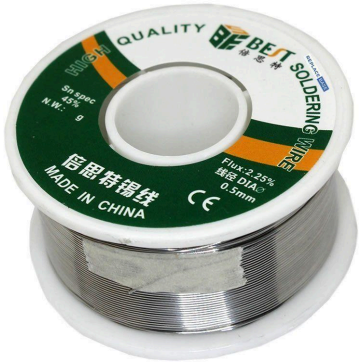


Figure 15. Solder wire **[10]** (the image is for demonstration purposes).

How Does a Soldering Iron Work?

The principle of operation is very simple: an electric current passes through a heating element inside the soldering iron, which heats the metal tip to a high temperature (usually between **200°C and 450°C**). When the hot tip is pressed against the joint and the solder wire, the tin melts, spreads over the joint and, after cooling for a few seconds, solidifies and creates a stable connection.

Main Types of Soldering Irons

- **Pencil (classic) soldering iron:** The simplest model, which plugs directly into a wall outlet. It has no advanced temperature control and constantly heats at full power. It is great for beginners and simpler repairs.

- **Soldering station (base):** Consists of a power unit with a controller and the handle of the soldering iron itself. It allows precise setting and maintenance of temperature, which is crucial for sensitive electronic components (such as microcomputers and SMD chips) so that they do not overheat.
- **Soldering gun:** Characterized by heating up extremely quickly (in just a few seconds) when the switch ("trigger") is pressed. It has high power and is used for thicker wires and coarser electrical work, but is not suitable for fine electronics.
- **Gas (cordless) soldering iron:** Runs on butane gas or batteries. It is very practical for work in the field where there is no access to the mains electricity.

Key Parts of a Soldering Iron

1. **Handle:** Made of a thermally insulating material (plastic, rubber or wood) to prevent heat from being transferred to the user's hand.
2. **Heating element:** Located inside the housing and converts electrical energy into heat.
3. **Soldering iron tip:** The most important part, which comes into direct contact with the tin. Tips can come in various shapes (chisel, conical, pointed) depending on the type of soldering, and can be replaced.
4. **Stand:** Serves for the safe placement of the hot soldering iron during work in order to avoid burns or fire.

Working tip: For the longevity of the soldering iron it is important to regularly clean its tip using a dedicated damp sponge or tip-cleaning wire, and to "tin" the tip (leave a thin layer of solder on it) before the tool is switched off in order to prevent oxidation.

Instructions

Wiring

Diagram

Component / Pin	**Connected to (Component / Pin)**	**Label / Value (WiringPi / Description)**
PCF8591		
VCC	Raspberry Pi Pin 4	5 V

GND	Raspberry Pi Pin 6	GND
SCL	Raspberry Pi Pin 5	9
SDA	Raspberry Pi Pin 3	8
MQ2 sensor module		
VCC	Raspberry Pi Pin 4	5 V
GND	Raspberry Pi Pin 6	GND
A0	PCF8591 - AIN0	Analog input
LCD display		
VSS	Raspberry Pi Pin 6	GND
VDD	Raspberry Pi Pin 4	5 V
V0	10 kΩ potentiometer - VOUT	Contrast
RS	Raspberry Pi Pin 37	25

R/W	Raspberry Pi Pin 6	GND
E	Raspberry Pi Pin 35	24
D0	-	-
D1	-	-
D2	-	-
D3	-	-
D4	Raspberry Pi Pin 33	23
D5	Raspberry Pi Pin 31	22
D6	Raspberry Pi Pin 29	21
D7	Raspberry Pi Pin 23	14
A	Raspberry Pi Pin 4	5 V
K	Raspberry Pi Pin 6	GND
10 kΩ potentiometer		

VCC	Raspberry Pi Pin 4	5 V
GND	Raspberry Pi Pin 6	GND
Active buzzer		
+ terminal	Raspberry Pi Pin 4	5 V (5VO)
- terminal	Raspberry Pi Pin 6	GND
SIG	Raspberry Pi Pin 36 (GPIO 16)	Activation signal
Transistor board		
IN1	Raspberry Pi Pin 11 (GPIO 17)	Signal input 1
IN2	Raspberry Pi Pin 13 (GPIO 27)	Signal input 2
IN3	Raspberry Pi Pin 15 (GPIO 22)	Signal input 3
VCC	Raspberry Pi Pin 4	5 V (5VO)
GND	Raspberry Pi Pin 9	GND

DC motor		
+ terminal	Raspberry Pi Pin 4	5 V (5VO)
- terminal	Raspberry Pi Pin 6	GND
SIG	Raspberry Pi Pin 12 (GPIO 18)	Signal (PWM)

Table 3. Wiring diagram.

Soldering the Transistors, Diodes and Wires onto the Perfboard

First, the voltage arriving from the microcomputer is too strong and would give too strong a current for our diodes. That is why we connect resistors. However, the resistors will reduce the current too much, so with the help of transistors we adjust the voltage so that the diodes activate when the signal arrives.

Diagram

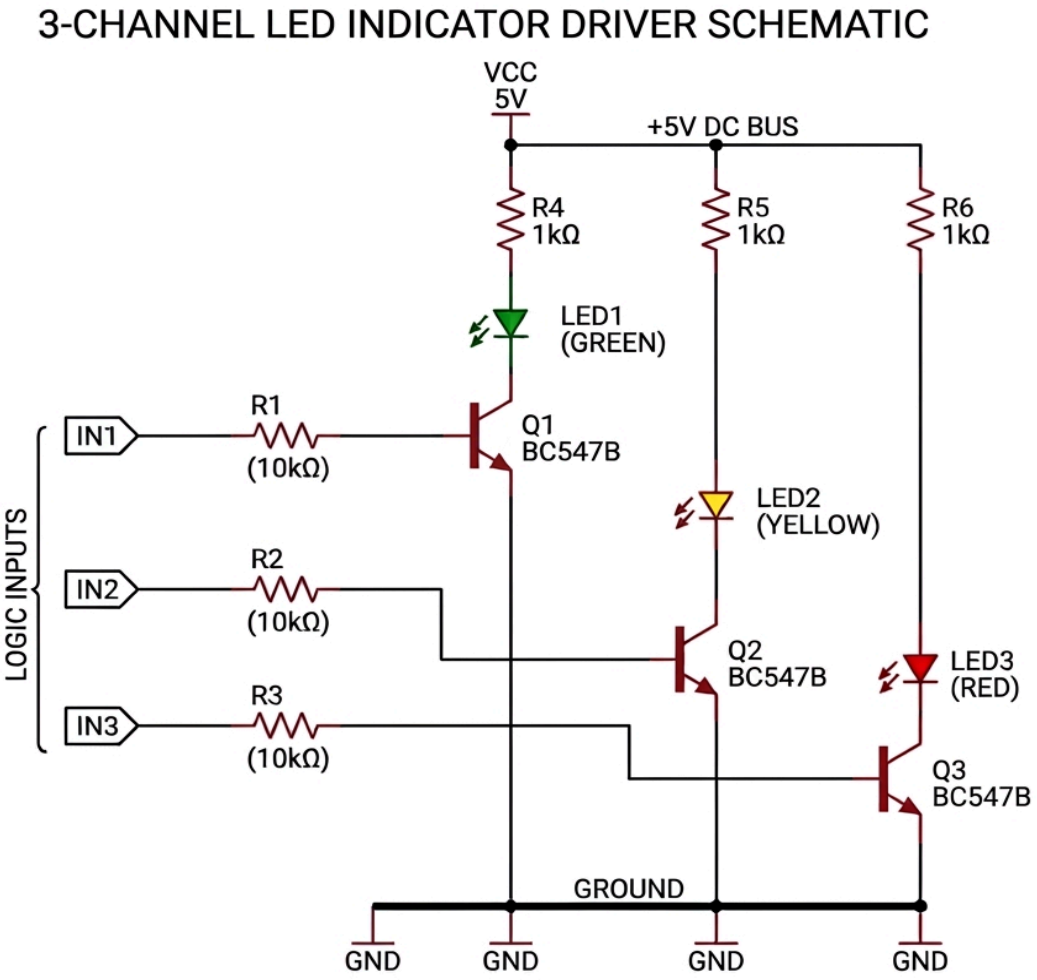


Figure 16. Driver diagram

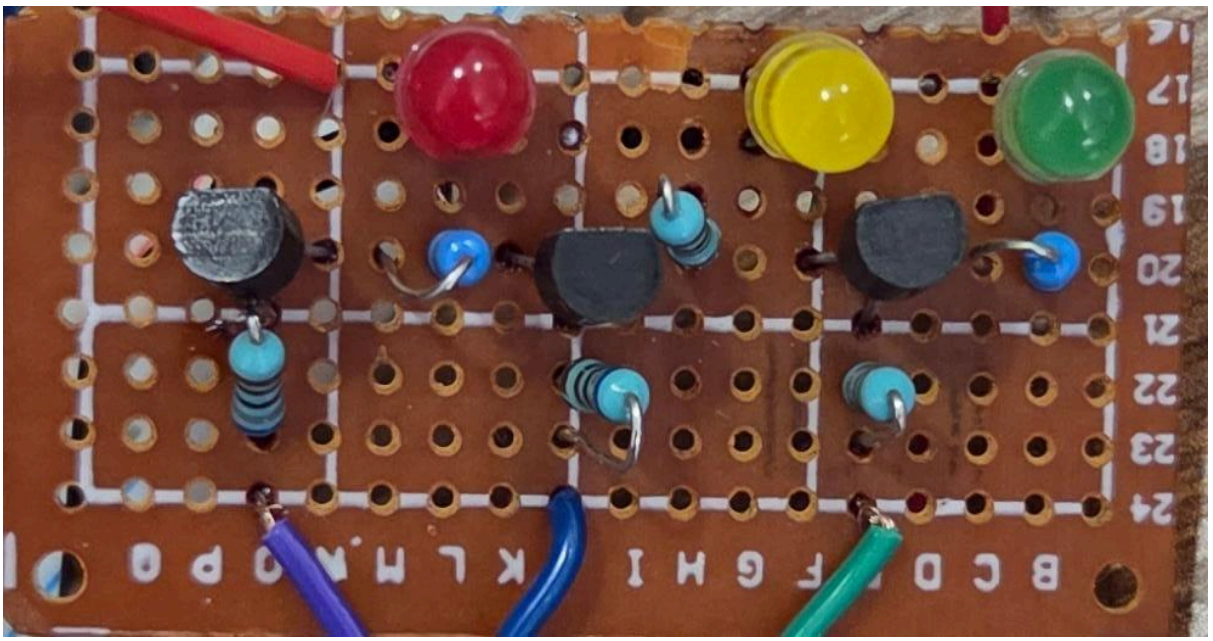


Figure 17. Actual appearance of the driver.

After wiring, you can move on to the code.
OPTIONAL: The MQ-2 has light indication (it has its own little lamp) when it registers pollution, and when it will light up can be set by turning its potentiometer. Our 3 LEDs offer better and richer information in any case.

Code

NOTE: The I2C bus must be enabled, as follows:

1. Open a terminal (Ctrl + Alt + t; or right-click and Open in terminal, or as a desktop application. It makes absolutely no difference).
2. Enter: sudo raspi-config
3. Then **Interface Options → I2C → Yes**.
4. (optional; for me it worked without this, and for sudo it didn't even ask for a password) sudo reboot, or through the graphical interface
   Or
1. Open a terminal
2. sudo raspi-config nonint do_i2c 1
3. the same as above

NOTE: You also need to download the lwiringPi library, as follows:

1. Open a terminal (anywhere and in any way)
2. Enter: sudo apt-get install wiringpi

This will probably not work. In that case:

1. Open a terminal (anywhere and in any way)
2. Enter:
3. cd /tmp

4. wget https://github.com/WiringPi/WiringPi/releases/download/3.2/wiringpi_3.2_arm64.deb
5. sudo dpkg -i wiringpi_3.2_arm64.deb
6.

If this does not work either:

1. Open a terminal (in any way)
2. Enter:
3. git clone https://github.com/WiringPi/WiringPi.git
4. cd WiringPi
5. ./build

On an already running Raspberry Pi 4 B microcomputer, save the following code in one of the folders:

```
#include <stdio.h>        /* printf, perror */
#include <stdlib.h>       /* exit */
#include <fcntl.h>        /* open */
#include <unistd.h>       /* read, write, close */
#include <sys/ioctl.h>    /* ioctl */
#include <linux/i2c-dev.h>/* I2C_SLAVE */
#include <wiringPi.h>     /* GPIO control - digitalWrite,
pinMode... */
#include <lcd.h>          /* LCD control - lcdInit, lcdPuts...
*/
#include <signal.h>       /* signal - catching Ctrl+C */

/* I2C address of the PCF8591 ADC converter - determined by
hardware (A0,A1,A2 = GND) */
#define ADRESA 0x48

/*
 * WiringPi pin numbering - DIFFERENT from GPIO numbering!
 * Use the "gpio readall" command to view all pins and their
numbers
 */
#define BAZ_pin    27  /* GPIO 16, physical pin 36 - buzzer
(active low: LOW=on) */
#define LED_ZELENA  0  /* GPIO 17, physical pin 11 - green LED
(clean air) */
#define LED_ZUTA    2  /* GPIO 27, physical pin 13 - yellow
LED (warning) */
```

```
#define LED_CRVENA  3  /* GPIO 22, physical pin 15 - red LED
(danger) */
#define MOTOR       1  /* GPIO 18, physical pin 12 - DC
motor/fan */

/* Global variables - must be global so that cleanup() can use
them */
int fd;   /* file descriptor for I2C communication with
/dev/i2c-1 */
int LCD;  /* handle returned by lcdInit(), used for all LCD
operations */

/*
 * cleanup() - function that is called automatically when the
user presses Ctrl+C
 * or when the program receives a SIGTERM signal (e.g. the
kill command)
 * Its role is to cleanly switch off all components before the
program exits,
 * so that they do not stay on after the program stops running
 */
void cleanup(int sig)
{
    lcdClear(LCD);                  /* clear the LCD screen */
    digitalWrite(LED_ZELENA, LOW);  /* turn off the green LED
*/
    digitalWrite(LED_ZUTA, LOW);    /* turn off the yellow LED
*/
    digitalWrite(LED_CRVENA, LOW);  /* turn off the red LED */
    digitalWrite(BAZ_pin, HIGH);    /* turn off the buzzer -
HIGH because it is active low */
    digitalWrite(MOTOR, LOW);       /* turn off the DC motor
*/
    close(fd);                      /* close the I2C
connection */
    exit(0);                        /* exit the program */
}

/*
 * setLed() - helper function for controlling all three LEDs
at once
 * Always takes values for all three LEDs at once (HIGH or
LOW)
```

```
 * This prevents more than one LED from lighting at the same
time
 */
void setLed(int zelena, int zuta, int crvena)
{
    digitalWrite(LED_ZELENA, zelena);
    digitalWrite(LED_ZUTA, zuta);
    digitalWrite(LED_CRVENA, crvena);
}

int main()
{
          int thresh1 = 45;         // threshold 1
    int thresh2 = 55;          // threshold 2
    /* Register the cleanup() function to be called when the
program is interrupted */
    signal(SIGINT, cleanup);   /* SIGINT  = Ctrl+C */
    signal(SIGTERM, cleanup);  /* SIGTERM = kill command */

    /* Initialization of the wiringPi library for controlling
GPIO pins */
    if (wiringPiSetup() == -1) exit(1);

    /* Set all pins as outputs (OUTPUT) */
    pinMode(BAZ_pin, OUTPUT);
    pinMode(LED_ZELENA, OUTPUT);
    pinMode(LED_ZUTA, OUTPUT);
    pinMode(LED_CRVENA, OUTPUT);
    pinMode(MOTOR, OUTPUT);

    /* Initial state - all components off when the program
starts */
    digitalWrite(BAZ_pin, HIGH);  /* HIGH = off (the buzzer is
active low) */
    digitalWrite(MOTOR, LOW);     /* motor off */
    setLed(LOW, LOW, LOW);        /* all three LEDs off */

    /*
     * LCD display initialization
     * lcdInit(rows, columns, bits, RS, E, D4, D5, D6, D7,
...)
     * We use 4-bit mode with pins: RS=25, E=24, D4=23, D5=22,
D6=21, D7=14
     */
```

```
    LCD = lcdInit(2, 16, 4, 25, 24, 23, 22, 21, 14, 0, 0, 0,
0);
    if (LCD == -1) { printf("LCD initialization failed!\n");
exit(2); }

    /* Opening the I2C bus as a file - Linux treats devices as
files */
    fd = open("/dev/i2c-1", O_RDWR);
    if (fd < 0) { perror("i2c open"); exit(1); }

    /* Setting the I2C address of the device we communicate
with (PCF8591 = 0x48) */
    if (ioctl(fd, I2C_SLAVE, ADRESA) < 0) { perror("ioctl");
exit(1); }

    /* PCF8591 configuration - the value 0x00 selects channel
AIN0 for reading */
    unsigned char config = 0x00;
    write(fd, &config, 1);

    /* Print the title on the first line of the LCD - done
only once, outside the loop */
    lcdPosition(LCD, 0, 0);
    lcdPuts(LCD, "Result:         ");

    /* Main loop - constantly reads the sensor and reacts to
changes */
    while (1)
    {
        unsigned char val;  /* variable for storing the value
from the sensor (0-255) */

        /*
         * PCF8591 works with a lag - it always returns the
value of the PREVIOUS conversion.
         * Solution: we read twice with a pause in between:
         *   1. first read - starts a new conversion, we
ignore the result
         *   2. delay(10)    - we wait 10 ms for the ADC to
finish the conversion
         *   3. second read - now we get the actual current
value
         */
```

```
        read(fd, &val, 1);  /* first read - discard, this is
the old value */
        delay(10);          /* wait for the ADC to finish the
conversion */
        read(fd, &val, 1);  /* second read - this is the
actual current value */

        /* Print the value to the terminal for monitoring and
debugging */
        printf("Value: %d\n", val);

        /*
         * Decision based on the sensor value - three danger
levels:
         *
         * LEVEL 1 - Clean air (val <= 45):
         *   The green LED is on, the buzzer and motor are off
         *
         * LEVEL 2 - Warning (46 <= val <= 55):
         *   The yellow LED lights up as a warning, the buzzer
and motor are still off
         *
         * LEVEL 3 - Danger (val > 55):
         *   The red LED is on, the buzzer sounds, the motor
starts and ventilates the room
         *   The system automatically drops to a lower level
as soon as the value falls below the threshold
         */
        if (val <= thresh1)
        {
            setLed(HIGH, LOW, LOW);       /* turn on green,
turn off the others */
            digitalWrite(BAZ_pin, HIGH);  /* turn off the
buzzer */
            digitalWrite(MOTOR, LOW);     /* turn off the
motor */
        }
        else if (val <= thresh2)
        {
            setLed(LOW, HIGH, LOW);       /* turn on yellow,
turn off the others */
            digitalWrite(BAZ_pin, HIGH);  /* buzzer still off
*/
```

```
            digitalWrite(MOTOR, LOW);     /* motor still off
*/
        }
        else
        {
            setLed(LOW, LOW, HIGH);       /* turn on red, turn
off the others */
            digitalWrite(BAZ_pin, LOW);   /* LOW = on (active
low buzzer) */
            digitalWrite(MOTOR, HIGH);    /* HIGH = turn on
the fan */
        }

        /*
         * Print the value on the second line of the LCD
         * "%-16d" - left-aligned number padded to 16
characters
         * This way we overwrite the old value without
lcdClear(), which flickers
         */
        lcdPosition(LCD, 0, 1);
        lcdPrintf(LCD, "%-16d", val);

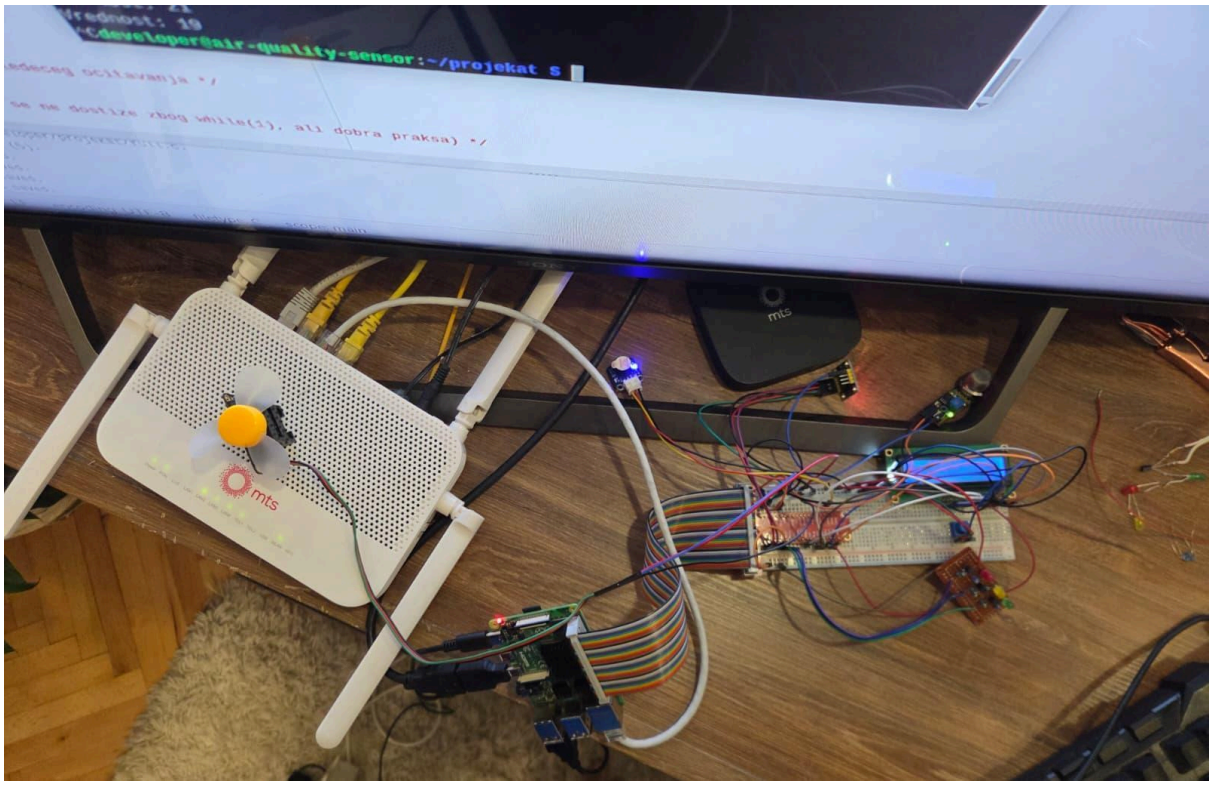
        delay(500);  /* pause of 500 ms before the next
reading */
    }

    close(fd);  /* close I2C (never reached because of
while(1), but good practice) */
    return 0;
}
```

And compile. For example, if you saved it as projekat.c, it looks like this:

1. Enter the root folder of the project (where the code is saved directly)
2. Open a terminal (IMPORTANT: this library must be installed! If it is not, an error will be reported!)
3. Enter: gcc projekat.c -o projekat -lwiringPi -lwiringPiDev

The finished project looks roughly like this:



Ideas for Extending the Project

- **Remote user notification (SMS / Email / Push notifications)** – Implementation of a service for immediately sending critical warnings and information about the system's alarm states to the user's mobile phone.
- **Centralized storage of logs and measurement history** – Creating a local or cloud database for permanent storage of the values read from the MQ2 sensor, which enables later statistical analysis of air quality.
- **Advanced web interface for monitoring and configuration** – Development of a responsive web application that gives the user insight into the current state of the

system in real time, a view of charts with historical data, and the ability to configure key parameters in software (critical trigger thresholds, measurement intervals, sampling frequency and the control logic of the actuators).

- **Integration of a local user interface** – Upgrading the display on the device itself (e.g. implementing a more advanced graphic display or a key matrix) for local output of detailed diagnostics about the system state and easier manual interaction.
- **Connecting to smart home ecosystems (IoT / MQTT)** – Integrating the system with central home servers via the MQTT protocol, which enables network coupling with other actuating devices, such as automatic activation of home air purifiers, smart extractor hoods or ventilation systems.
- **Automatic self-diagnostics and software calibration of the sensor** – Development of an algorithm that dynamically tracks the base value (*baseline*) of the MQ2 sensor's resistance in clean-air conditions, thereby compensating for the natural degradation of the sensing material over time and sending a notification when physical replacement of the component is necessary.
- **Switching to a backup power source (UPS module)** – Adding a protective hardware HAT with battery support that ensures continuous operation of the primary safety functions (sensor and buzzer) and the sending of urgent notifications even in scenarios of complete loss of power from the mains.
- **Voice warnings using Text-to-Speech (TTS) technology** – Using the audio output of the development platform to generate clear verbal warnings and voice messages in real time, thereby precisely communicating to the user the nature and level of the detected danger.

Potential Problems

Note on the time delay during AD conversion:

When implementing the software code for reading values from the MQ2 sensor, special attention must be paid to the internal conversion time of the analog signal required by the PCF8591 AD/DA converter. If readings inside the loop are performed successively without an adequate time interval, the converter, due to the time delay, fails to complete the conversion cycle, which results in incorrect (default*default*) values being sent to the system. This problem is effectively solved by introducing a minimal software delay of delay(10); immediately after each reading, which gives the converter enough time to stabilize and accurately digitize the signal.

Note on the behavior of the audible signaling device (buzzer):

In practice, it may happen that an active audible signaling device emits a continuous tone even when the executable program on the development platform is not running. Such uncontrolled activation is a consequence of the fact that, in the initial state, before the software is initialized, the GPIO pins may be in an undefined logic state. Since a buzzer driven by a low logic level (LOW signal) is used, any drop in potential on the corresponding

pin closes the circuit and activates the audible signaling, which requires hardware or software definition of the initial pin state (with a*pull-up* configuration).

References

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